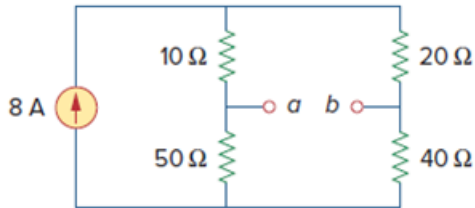


Problem

Use *PSpice* or *MultiSim* to find the Thevenin equivalent circuit at terminals *a-b* of the circuit in Fig. 4.125.

Figure. 4.125:



Step-by-step solution

Step 1 of 3

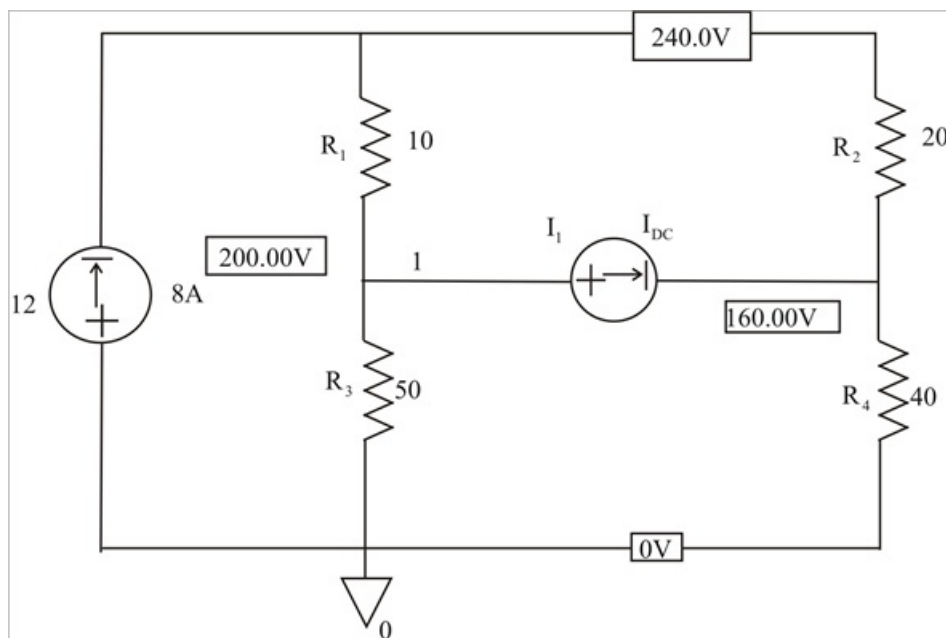
The schematic is shown below. We label nodes *a* and *b* as 1 and 2 respectively. We perform a DC sweep on I1. In the trace or add menu type $V(1) - V(2)$ which will result in the plot below.

From the plot,

$$V_{TH} = 40V \text{ [Zero Intercept] .}$$

$$R_{TH} = \frac{(40 - 17.5)}{1} = 22.5 \text{ Ohms .[slope]}$$

Step 2 of 3



Step 3 of 3

